

A.3 Witt-vector algebra

A.3.1 Ring structure and ghost coordinates

For $n \geq 0$, the n -th **Witt polynomial** in the variables $X_0, \dots, X_n$ is

$$w_n(X_0, \dots, X_n) = \sum_{j=0}^n p^j X_j^{p^{n-j}} = X_0^{p^n} + pX_1^{p^{n-1}} + \dots + p^n X_n.$$

Theorem 77 (Witt; [Ser79, II, §6, Theorem 6], [Rab14, §1]). *There exists a unique family of polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$, $n \geq 0$, such that*

$$\begin{aligned} w_n(S_0, \dots, S_n) &= w_n(X_0, \dots, X_n) + w_n(Y_0, \dots, Y_n), \\ w_n(P_0, \dots, P_n) &= w_n(X_0, \dots, X_n) \cdot w_n(Y_0, \dots, Y_n). \end{aligned}$$

for all n . For any commutative ring A , the operations

$$\vec{a} \oplus \vec{b} := (S_n(\vec{a}; \vec{b}))_{0 \leq n \leq r-1}, \quad \vec{a} \odot \vec{b} := (P_n(\vec{a}; \vec{b}))_{0 \leq n \leq r-1}$$

make the set A^r into a commutative ring $W_r(A)$, the ring of *p -typical Witt vectors of length r over A* , with zero $(0, \dots, 0)$ and unit $(1, 0, \dots, 0)$. Additive inverses are likewise given by universal integral polynomials; we write $\ominus$ for Witt subtraction. The **ghost map**

$$w: W_r(A) \longrightarrow A^r, \quad \vec{a} \longmapsto (w_0(\vec{a}), w_1(\vec{a}), \dots, w_{r-1}(\vec{a}))$$

is a ring homomorphism to the product ring A^r ; it is injective when p is a non-zero-divisor in A , and bijective when $p \in A^\times$.

Equivalently, the universal Witt addition and inverse polynomials make $\mathbb{A}_{\mathbb{Z}}^r$ an affine commutative group scheme. After base change to $\mathbb{F}_p$, this is the additive Witt-vector group scheme $\mathbb{W}_r$ used in Section A.4.

We index the components of a Witt vector $\vec{a} = (a_0, \dots, a_{r-1})$ by $0, \dots, r-1$. For $r = 1$ the polynomials reduce to $S_0 = X_0 + Y_0$ and $P_0 = X_0 Y_0$, so $W_1(A) = A$ as a ring.

Since the ring operations are given by universal polynomials with integer coefficients, W_r is a functor: a ring homomorphism $\sigma: A \rightarrow B$ induces the ring homomorphism

$$W_r(\sigma): W_r(A) \rightarrow W_r(B), \quad W_r(\sigma)(a_0, \dots, a_{r-1}) = (\sigma a_0, \dots, \sigma a_{r-1}),$$

acting *componentwise*. Two consequences of this trivial-looking observation are used repeatedly and deserve emphasis: a ring automorphism σ of A commutes with $\oplus$ and $\ominus$ on $W_r(A)$, and a Witt vector $\vec{a} \in W_r(A)$ satisfies $W_r(\sigma)(\vec{a}) = \vec{a}$ if and only if *each component a_n is σ -invariant*. In particular, for a group Γ acting on A by ring automorphisms,

$$W_r(A)^\Gamma = W_r(A^\Gamma) \quad \text{inside } W_r(A). \quad (\text{A.6})$$

The uniqueness statement in Theorem 77 is an instance of a general principle that we invoke several times, so we record it explicitly.

Lemma 78 (ghost principle). *Let Φ and Ψ be two operations on Witt vectors given by universal polynomials with integer coefficients in the components of their arguments. If $w \circ \Phi = w \circ \Psi$ holds over the polynomial ring $\mathbb{Z}[X_0, X_1, \dots]$ on the components of the arguments, then $\Phi = \Psi$ over every commutative ring.*